

ASSESSING THE TREATMENT EFFECT ¹HETEROGENEITY WITH A LATENT VARIABLE ²Yunjian Yin^{1,2}, Lan Liu^{2,*}, and Zhi Geng¹ ³¹*School of Mathematical Sciences, Peking University, Beijing 100871, China* ⁴²*School of Statistics, University of Minnesota, Minneapolis, Minnesota 55455, USA*Appendix A Proof of Formulas ⁵Proof of Formulas (F1) and (F2) ⁶

$$\begin{aligned}
& \text{TBR}_c(x) = P(Y_1 - Y_0 > c | X = x) \\
&= P((\alpha_{1,0} - \alpha_{0,0}) + (\alpha_{1,1} - \alpha_{0,1})^T x + (\alpha_{1,2} - \alpha_{0,2})U + (\alpha_{1,3} - \alpha_{0,3})^T xU + (\epsilon_1 - \epsilon_0) > c) \\
&= P\left(\frac{\epsilon_0 - \epsilon_1}{\sqrt{\sigma_0^2 + \sigma_1^2}} < \frac{(\alpha_{1,0} - \alpha_{0,0}) + (\alpha_{1,1} - \alpha_{0,1})^T x + (\alpha_{1,2} - \alpha_{0,2})U + (\alpha_{1,3} - \alpha_{0,3})^T xU - c}{\sqrt{\sigma_0^2 + \sigma_1^2}}\right) \\
&= \int \Phi\left(\frac{(\alpha_{1,0} - \alpha_{0,0}) + (\alpha_{1,1} - \alpha_{0,1})^T x + (\alpha_{1,2} - \alpha_{0,2})u + (\alpha_{1,3} - \alpha_{0,3})^T xu - c}{\sqrt{\sigma_0^2 + \sigma_1^2}}\right) f_U(u) du \\
&= \int \Phi\left((w_1 + w_2 u)/w_3\right) f_U(u) du \\
&= \int \int_{-\infty}^{(w_1 + w_2 u)/w_3} \frac{1}{\sqrt{2\pi}} \exp(-s^2/2) f_U(u) ds du \\
&= \int \int_{-\infty}^0 \frac{1}{2\pi} \exp\left[-\frac{1}{2w_3^2} \left\{ (w_2^2 + w_3^2) \left(u + \frac{w_2(w_3 s + w_1)}{w_2^2 + w_3^2}\right)^2 + \frac{w_3^2(w_3 s + w_1)^2}{w_2^2 + w_3^2} \right\}\right] ds du \\
&= \int_{-\infty}^0 \frac{1}{\sqrt{2\pi}} \sqrt{\frac{w_3^2}{w_2^2 + w_3^2}} \exp\left\{-\frac{(w_3 s + w_1)^2}{2(w_2^2 + w_3^2)}\right\} ds \\
&= \Phi\left(\frac{w_1}{\sqrt{w_2^2 + w_3^2}}\right),
\end{aligned}$$

where $f_U(\cdot)$ is the density functions of U , $w_1 = (\alpha_{1,0} - \alpha_{0,0}) + (\alpha_{1,1} - \alpha_{0,1})^T x - c$, $w_2 = (\alpha_{1,2} - \alpha_{0,2}) + (\alpha_{1,3} - \alpha_{0,3})^T x$, $w_3 = \sqrt{\sigma_0^2 + \sigma_1^2}$. Similarly, we can derive the form for $\text{THR}_c(x)$. ⁸

Proof of Formulas (F3) and (F4) ⁹

Let $K(\alpha_t, x, u) = \alpha_{t,0} + \alpha_{t,1}^T x + \alpha_{t,2} u + \alpha_{t,3}^T x u$, we have 1

$$\begin{aligned} \text{TBR}(x) &= \int \{1 - \Phi(K(\alpha_0, x, u))\} \Phi(K(\alpha_1, x, u)) f_U(u) du \\ &= \int \left\{ \int_{K(\alpha_0, x, u)}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-s_0^2/2) ds_0 \right\} \left\{ \int_{-\infty}^{K(\alpha_1, x, u)} \frac{1}{\sqrt{2\pi}} \exp(-s_1^2/2) ds_1 \right\} f_U(u) du \\ &= \int \int_{-\infty}^0 \int_0^{\infty} \frac{1}{(2\pi)^{3/2}} \exp \left\{ -\frac{(s_0 + K(\alpha_0, x, u))^2 + (s_1 + K(\alpha_1, x, u))^2 + u^2}{2} \right\} ds_0 ds_1 du. \end{aligned} \quad \text{2}$$

Let $K_1(\alpha_t, x) = \alpha_{t,0} + \alpha_{t,1}^T x$, $K_2(\alpha_t, x) = \alpha_{t,2} + \alpha_{t,3}^T x$, thus $K(\alpha_t, x, u) = K_1(\alpha_t, x) + u K_2(\alpha_t, x)$. 3

Then 4

$$\begin{aligned} & \{s_0 + K(\alpha_0, x, u)\}^2 + \{s_1 + K(\alpha_1, x, u)\}^2 + u^2 \\ &= \{1 + K_2(\alpha_0, x)^2 + K_2(\alpha_1, x)^2\} u^2 + 2\{(s_0 + K_1(\alpha_0, x))K_2(\alpha_0, x) + (s_1 + K_1(\alpha_1, x))K_2(\alpha_1, x)\}u \\ & \quad + \{s_0 + K_1(\alpha_0, x)\}^2 + \{s_1 + K_1(\alpha_1, x)\}^2 \\ &= \{1 + K_2(\alpha_0, x)^2 + K_2(\alpha_1, x)^2\} \left\{ u + \frac{(s_0 + K_1(\alpha_0, x))K_2(\alpha_0, x) + (s_1 + K_1(\alpha_1, x))K_2(\alpha_1, x)}{1 + K_2(\alpha_0, x)^2 + K_2(\alpha_1, x)^2} \right\}^2 \\ & \quad + \frac{1}{1 + K_2(\alpha_0, x)^2 + K_2(\alpha_1, x)^2} \left[\{s_0 + K_1(\alpha_0, x)\}^2 \{1 + K_2(\alpha_1, x)^2\} \right. \\ & \quad \quad \quad \left. + \{s_1 + K_1(\alpha_1, x)\}^2 \{1 + K_2(\alpha_0, x)^2\} \right. \\ & \quad \quad \quad \left. - 2\{s_0 + K_1(\alpha_0, x)\}K_2(\alpha_0, x)\{s_1 + K_1(\alpha_1, x)\}K_2(\alpha_1, x) \right]. \end{aligned} \quad \text{5}$$

So 6

$$\text{TBR}(x) = \int_0^{\infty} \int_{-\infty}^0 \frac{1}{(2\pi)S} \exp \left(-\frac{F}{2} \right) ds_0 ds_1, \quad \text{7} \quad (\text{A1.1})$$

where $S^2 = 1 + K_2(\alpha_0, x)^2 + K_2(\alpha_1, x)^2$, 8

$$\begin{aligned} F &= \left[\{s_0 + K_1(\alpha_0, x)\}^2 \{1 + K_2(\alpha_1, x)^2\} + \{s_1 + K_1(\alpha_1, x)\}^2 \{1 + K_2(\alpha_0, x)^2\} \right. \\ & \quad \left. - 2\{s_0 + K_1(\alpha_0, x)\}K_2(\alpha_0, x)\{s_1 + K_1(\alpha_1, x)\}K_2(\alpha_1, x) \right] / S^2 \\ &= \left\{ (s_0, s_1) - \mu \right\} \Sigma^{-1} \left\{ (s_0, s_1) - \mu \right\}^T, \end{aligned} \quad \text{9}$$

$$\mu = (-K_1(\alpha_0, x), -K_1(\alpha_1, x)), \quad \text{10}$$

$$\Sigma = \begin{pmatrix} 1 + K_2(\alpha_0, x)^2 & K_2(\alpha_0, x)K_2(\alpha_1, x) \\ K_2(\alpha_0, x)K_2(\alpha_1, x) & 1 + K_2(\alpha_1, x)^2 \end{pmatrix}. \quad 2$$

Thus, $\text{TBR}(x) = \Phi_2((0, \infty), (-\infty, 0); \mu, \Sigma)$, where $\Phi_2(A_0, A_1; \mu, \Sigma)$ is the distribution function of bivariate normal vector with mean μ , covariance matrix Σ and integral region $A_0 \times A_1$. Similarly, we can derive the form for $\text{THR}(x)$. 3

Relationship of $\text{ATE}(x)$, $\text{TBR}_c(x)$ and $\text{THR}_c(x)$ 4

Note for any random variable Z , we have 5

$$E(Z) = \int_0^\infty \{1 - F_Z(z)\}dz - \int_{-\infty}^0 F_Z(z)dz, \quad 6$$

where F_Z is the cumulative distribution function of Z . Thus, 7

$$\begin{aligned} \text{ATE}(x) &= E(Y_1 - Y_0 | X = x) \\ &= \int_0^\infty \{1 - F_{Y_1 - Y_0 | x}(c)\}dc - \int_{-\infty}^0 F_{Y_1 - Y_0 | x}(c)dc \\ &= \int_0^\infty \text{TBR}_c(x)dc - \int_{-\infty}^0 \{1 - \text{TBR}_c(x)\}dc \\ &= \int_0^\infty \text{TBR}_c(x)dc - \int_{-\infty}^0 \text{THR}_{-c}(x)dc \\ &= \int_0^\infty \{\text{TBR}_c(x) - \text{THR}_c(x)\}dc, \end{aligned} \quad 8$$

where the penultimate step holds since $Y_1 - Y_0$ is continuous. 9

Appendix B Proof of Theorem 1 10

Instead of proving Theorem 1 directly, we first provide sufficient and necessary identification conditions of $(g_t(X); h_t(X))$ in the general models (3) and (4). 11

Theorem B.1. *Under Assumption 2,* 12

(i) When the outcome is continuous, if the following model (A2.2) holds for $t=0,1$, **1**

$$\begin{cases} Y_t = g_t(X) + U h_t(X) + \epsilon_t, \\ \epsilon_t \perp (X, U), \epsilon_t \sim N(0, \sigma_t^2), U \sim N(0, 1), h_t(0) > 0, \end{cases} \quad (\text{A2.2}) \quad \text{2}$$

then the following Condition A is the sufficient and necessary condition to identify $(g_0(X), h_0(X), \sigma_0^2, g_1(X), h_1(X), \sigma_1^2)$ **3**

Condition A. $h_t(X)$ belongs to the family $\mathcal{S}(X)$ for $t = 0, 1$, where **4**

$$\mathcal{S}(X) = \{h(X) : h(X) \text{ can be identified if } h(X)h'(X) \text{ is known.}\} \quad \text{5}$$

(ii) When the outcome is continuous, if the following model (A2.3) holds for $t=0,1$, **6**

$$\begin{cases} Y_t^* = g_t(X) + U h_t(X) + \epsilon_t, \\ Y_t = I(Y_t^* > 0), \\ \epsilon_t \perp (X, U), \epsilon_t \sim N(0, \sigma_t^2), U \sim N(0, 1), h_t(0) > 0, \end{cases} \quad (\text{A2.3}) \quad \text{7}$$

then the following Condition B is the sufficient and necessary condition to identify $(g_0(X), h_0(X), g_1(X), h_1(X))$. **8**

Condition B. $(g_t(X), h_t(X))$ belongs to the family $(\mathcal{S}_1(X), \mathcal{S}_2(X))$ for $t = 0, 1$, **9**

where **10**

$$\begin{aligned} & (\mathcal{S}_1(X), \mathcal{S}_2(X)) \\ &= \left\{ (g(X; \alpha_1), h(X; \alpha_2)) \mid (\alpha_1, \alpha_2) \in \mathcal{A}, \forall (\alpha_1, \alpha_2) \neq (\beta_1, \beta_2) \in \mathcal{A}, \frac{g(X; \alpha_1)}{\sqrt{1 + h^2(X; \alpha_2)}} \neq \frac{g(X; \beta_1)}{\sqrt{1 + h^2(X; \beta_2)}} \right\}. \end{aligned} \quad \text{11}$$

Proof. **12**

(i) Since $E[Y|X, T = t] = E[Y_t|X] = g_t(X)$, we can identify $g_t(X)$ and we have **13**

$$(Y - g_t(X)) \mid (X, T = t) \sim N(0, h_t^2(X) + \sigma_t^2). \quad \text{14}$$

Thus $A_t(X) = h_t^2(X) + \sigma_t^2$ can also be identified, so is $A'_t(X) = h_t(X)h'_t(X)$. **15**

Next we show that Condition A is sufficient and necessary to identify $h_t(x), t = 0, 1$. **16**

It is easy to see that if $h_t(X)$ belongs to $\mathcal{S}(X)$, then $h_t(X)$ is also identified. On the

other hand, if $h_t(X)$ does not belong to $\mathcal{S}(X)$, then $h_t(X)$ can not be decided uniquely 1 from $h_t(X)h'_t(X)$. Besides, knowing $h_t(X)h'_t(X)$ is equivalent to knowing $h_t^2(X)$ up to a constant, i.e., $h_t^2(X_1) - h_t^2(X_2)$ for all X_1, X_2 . Note that $(Y - g_t(X)) \Big| (X, T = t) \sim N(0, h_t^2(X) + \sigma_t^2)$, the distribution of $Y - g_t(X)$ condition on $(X, T = t)$ is determined by the variance, so all the information we have about $h_t(X)$ is $h_t^2(X) + \sigma_t^2$, which is the same as knowing $h_t^2(X_1) - h_t^2(X_2)$ for all X_1, X_2 . Thus, we can not identify $h_t(X)$. So the sufficient and necessary condition is that $h_t(X) \in \mathcal{S}(X)$ for $t = 0, 1$.

(ii) Since $P(Y = 1|X, U, T = t) = \Phi(g_t(X) + Uh_t(X))$, we have 2

$$P(Y = 1|X, T = t) = \Phi\left(\frac{g_t(X)}{\sqrt{1 + h_t^2(X)}}\right), \quad \text{3}$$

It is easy to see that $(g_0(X), h_0(X), g_1(X), h_1(X))$ can be identified if and only if the 4

Condition B holds. 5 □

The identification of heterogeneous treatment effects given in Theorem 1 follows from the 6 following corollaries.

Corollary 1. When $h(X) = h(X; \eta) = \eta_0 + \eta_1^T X$, where $\eta = (\eta_0, \eta_1^T)^T$, $\eta_1 = (\eta_{1,1}, \dots, \eta_{1,p})^T$ 7 and $\eta_0 > 0$, we have $h(X) \in \mathcal{S}$.

Proof. Since $h(X)h'(X) = (\eta_0 + \eta_1^T X)\eta_1 = \eta_0\eta_1 + \eta_1\eta_1^T X$, we can identify $(\eta_0\eta_1, \eta_1\eta_1^T)$ if 8 $h(X)h'(X)$ is known. Besides, $h(0) = \eta_0 > 0$, so the sign of every component of η_1 can be determined since we know $\eta_0\eta_1$. Then η_1 can be identified since we know the diagonal elements of $\eta_1\eta_1^T$. Then η_0 can also be identified from $\eta_0\eta_1$. Thus (η_0, η_1) is identifiable, so is $h(X)$. This completes the proof of the part (i) in Theorem 1. □

We impose the following regularity condition on $\mathcal{X}$ which is the domain of X . 9

Condition C. *There exists linear independent $(\tau_1, \dots, \tau_p) \subset \mathcal{X}$, where $\mathcal{X}$ is the domain of X ,* 1
s.t. $P(Y = 1|X = \tau_i) = 0, i = 1, \dots, p$.

Corollary 2. *When $g(X) = g(X; \alpha) = \alpha_0 + \alpha_1^T X$, $h(X) = h(X; \alpha) = \alpha_2 + \alpha_3^T X$ with $(\alpha_0, \alpha_1) \neq$ 2
 $0, \alpha_2 > 0, \alpha_3 \neq 0$, where $\alpha = (\alpha_0, \alpha_1^T, \alpha_2, \alpha_3^T)^T, \alpha_1 = (\alpha_{1,1}, \dots, \alpha_{1,p})^T, \alpha_3 = (\alpha_{3,1}, \dots, \alpha_{3,p})^T$,
 if the Condition C holds, we have $\{g(X), h(X)\} \in \{\mathcal{S}_1(X), \mathcal{S}_2(X)\}$.*

Proof. It is enough to show that if $\alpha = (\alpha_0, \alpha_1^T, \alpha_2, \alpha_3^T)^T, \beta = (\beta_0, \beta_1^T, \beta_2, \beta_3^T)^T$ satisfy: 3

$$\frac{\alpha_0 + \alpha_1^T X}{\sqrt{1 + (\alpha_2 + \alpha_3^T X)^2}} = \frac{\beta_0 + \beta_1^T X}{\sqrt{1 + (\beta_2 + \beta_3^T X)^2}}, \quad \forall X \in \mathcal{X}, \quad \text{4} \quad (\text{A2.4})$$

then $\alpha = \beta$. To keep the same signs on both sides, the following two subsets of a hyperplane 5
 (H_0, H_1) must be the same,

$$H_0 = \{X \in \mathcal{X} | \alpha_0 + \alpha_1^T X = 0\}, \quad H_1 = \{X \in \mathcal{X} | \beta_0 + \beta_1^T X = 0\}, \quad \text{6}$$

since there exists linear independent $(\tau_1, \dots, \tau_p) \subset \mathcal{X}$ such that $P(Y = 1|X = \tau_i) = 0.5, i =$ 7
 $1, \dots, p$, thus, the following two hyperplane $(\tilde{H}_0, \tilde{H}_1)$ must be the same,

$$\tilde{H}_0 = \{X \in \mathbb{R}^p | \alpha_0 + \alpha_1^T X = 0\}, \quad \tilde{H}_1 = \{X \in \mathbb{R}^p | \beta_0 + \beta_1^T X = 0\}, \quad \text{8}$$

which means $(\alpha_0, \alpha_1^T) = k(\beta_0, \beta_1^T)$, and $k \geq 0$ since the signs on the two sides of equations 9
 (A2.4) must be the same. And $(\alpha_0, \alpha_1) \neq 0$ exclude the case $k = 0$. Thus from equation (A2.4)
 we have

$$k^2 = \frac{1 + (\alpha_2 + \alpha_3^T X)^2}{1 + (\beta_2 + \beta_3^T X)^2}. \quad \text{10}$$

By arranging the equation above we have 11

$$X^T(\alpha_3 \alpha_3^T - k^2 \beta_3 \beta_3^T)X + 2(\alpha_2 \alpha_3^T - k^2 \beta_2 \beta_3^T)X + 1 + \alpha_2^2 - k - k\beta_2^2 = 0. \quad \text{12}$$

So **1**

$$\alpha_3 \alpha_3^T - k^2 \beta_3 \beta_3^T = 0, \quad \mathbf{2} \quad (\text{A2.5a})$$

$$\alpha_2 \alpha_3^T - k^2 \beta_2 \beta_3^T = 0, \quad \mathbf{3} \quad (\text{A2.5b})$$

$$1 + \alpha_2^2 - k^2 - k^2 \beta_2^2 = 0. \quad \mathbf{4} \quad (\text{A2.5c})$$

With a little abuse of notation, we use 0 to denote not only the number 0 but also the matrix **5** and vector of 0 in (A2.5a) and (A2.5b) respectively. Take the (i, i) element of (A2.5a) and the i -th component of (A2.5b), with a little arrangement we have

$$\alpha_{3i}^2 = k^2 \beta_{3i}^2, \quad \mathbf{6} \quad (\text{A2.5d})$$

$$\alpha_2 \alpha_{3i} = k^2 \beta_2 \beta_{3i}, \quad \mathbf{7} \quad (\text{A2.5e})$$

$$\alpha_2^2 = k^2 + k^2 \beta_2^2 - 1. \quad \mathbf{8} \quad (\text{A2.5f})$$

Note $(\text{A2.5d}) \cdot (\text{A2.5f}) - (\text{A2.5e})^2 = k^2 \beta_{3i}^2 (k^2 - 1) = 0$, since $k > 0$ we have $k = 1$. And since $\alpha_2, \beta_2 \geq 0$, from (A2.5c) we have $\alpha_2 = \beta_2$, then from (A2.5b) we have $\alpha_3 = \beta_3$. Thus, $\alpha = \beta$. This completes the proof of part (ii) in Theorem 1. $\square$ **9**

Appendix C Non-identification without interaction term **10**

between X and U

Theorem C.1. *Under the same assumptions as in Theorem B.1, **11***

- (i) *If there is no interaction term between X and U in model (A2.2), i.e., $h_t(X) = h_t$ is a **12** constant, the $(\text{TBR}_c(x), \text{THR}_c(x))$ can not be identified for any $c \neq \pm E[Y_1 - Y_0]$.*
- (ii) *If there is no interaction term between X and U in model (A2.3), i.e., $h_t(X) = h_t$ is a constant, the $(\text{TBR}(x), \text{THR}(x))$ can not be identified for any $(g_0(x), g_1(x)) \neq (0, 0)$.*

Proof. 1

(i) We have 2

$$Y \Big| \left(X, T = t \right) \sim N \left(g_t(X), h_t^2 + \sigma_t^2 \right). \quad \textcolor{red}{3}$$

Since $P(Y, X, T) = P(Y|X, T)P(X, T)$ and $P(X, T)$ is not related to the parameters in 4 the model, we can only identify $g_t(X)$ and $h_t^2 + \sigma_t^2$ for $t = 0, 1$. Since h_t^2 is a constant, we can no longer separate h_t and σ_t^2 from $(h_t^2 + \sigma_t^2)$ without further assumptions, i.e., (h_t, σ_t^2) can not be identified. Additionally, we have

$$(Y_0, Y_1) \Big| X = x \sim N(\mu(x), \Sigma(x)), \quad \textcolor{red}{5}$$

where 6

$$\mu(x) = (g_0(x), g_1(x)), \quad \Sigma(x) = \begin{pmatrix} h_0^2 + \sigma_0^2 & h_0 h_1 \\ h_0 h_1 & h_1^2 + \sigma_1^2 \end{pmatrix}. \quad \textcolor{red}{7}$$

Thus, 8

$$(Y_1 - Y_0) | X = x \sim N \left(g_1(x) - g_0(x), (h_0^2 + \sigma_0^2) + (h_1^2 + \sigma_1^2) - 2h_0 h_1 \right). \quad \textcolor{red}{9}$$

Since $h_t^2 + \sigma_t^2$ can be identified while (h_t^2, σ_t^2) can not, the joint distribution of (Y_0, Y_1) 10 given $X = x$ can not be identified, so is the distribution of $Y_1 - Y_0$ given $X = x$.

Since $\text{TBR}_c(x) = P(Y_1 - Y_0 > c | X = x)$ and $Y_1 - Y_0$ given $X = x$ is normally distributed 11 with mean identified and variance unidentified, so $\text{TBR}_c(x)$ is unidentified if $c \neq E[Y_1 - Y_0]$. Similarly, $\text{THR}_c(x)$ is unidentified if $c \neq -E[Y_1 - Y_0]$.

(ii) Since 12

$$P(Y = 1 | X, T = t) = \Phi \left(\frac{g_t(X)}{\sqrt{1 + h_t^2}} \right), \quad \textcolor{red}{13}$$

we can only identify $g_t(X)/\sqrt{1 + h_t^2}$ in the model with the numerator and denominator 14

unseparate, which means $(g_t(X), h_t^2)$ can not be identified. Additionally, we have ¹

$$\begin{aligned} \text{TBR}(x) &= P(Y_0 = 0, Y_1 = 1 | X = x) \\ &= \int_{-\infty}^0 \int_0^{\infty} \frac{1}{2\pi|\Sigma_b|^{1/2}} \exp \left\{ -\frac{1}{2}((s_0, s_1) - \mu_b)\Sigma_b^{-1}((s_0, s_1) - \mu_b) \right\} ds_0 ds_1, \end{aligned} \quad 2$$

where ³

$$\mu_b = (-g_0(x), -g_1(x)), \quad \Sigma_b = \begin{pmatrix} 1 + h_0^2 & h_0 h_1 \\ h_0 h_1 & 1 + h_1^2 \end{pmatrix}. \quad 4$$

Let $(t_0 = s_0/\sqrt{1+h_0^2}, t_1 = s_1/\sqrt{1+h_1^2})$, we have ⁵

$$\begin{aligned} \text{TBR}(x) &= P(Y_0 = 0, Y_1 = 1 | X = x) \\ &= \int_{-\infty}^0 \int_0^{\infty} \frac{1}{2\pi|\tilde{\Sigma}_b|^{1/2}} \exp \left\{ -\frac{1}{2}((t_0, t_1) - \tilde{\mu}_b)\tilde{\Sigma}_b^{-1}((t_0, t_1) - \tilde{\mu}_b) \right\} dt_0 dt_1, \end{aligned} \quad 6$$

where ⁷

$$\tilde{\mu}_b = (-g_0(x)/\sqrt{1+h_0^2}, -g_1(x)/\sqrt{1+h_1^2}), \quad \tilde{\Sigma}_b = \begin{pmatrix} 1 & \frac{h_0 h_1}{\sqrt{1+h_0^2}\sqrt{1+h_1^2}} \\ \frac{h_0 h_1}{\sqrt{1+h_0^2}\sqrt{1+h_1^2}} & 1 \end{pmatrix}. \quad 8$$

So $\tilde{\mu}_b$ is identified while $\tilde{\Sigma}_b$ not. Thus, we can easily conclude that $\text{TBR}(x)$ can not be identified when $(g_0(x), g_1(x)) \neq (0, 0)$, so is $\text{THR}(x)$ and the joint distribution of (Y_0, Y_1) given $X = x$. $\square$ ⁹

Appendix D Proof of Theorem 2 ¹⁰

Proof. The estimator $\hat{\theta} = (\hat{\alpha}_{0,0}, \hat{\alpha}_{0,1}^T, \hat{\alpha}_{0,2}, \hat{\alpha}_{0,3}^T, \hat{\sigma}_0^2, \hat{\alpha}_{1,0}, \hat{\alpha}_{1,1}^T, \hat{\alpha}_{1,2}, \hat{\alpha}_{1,3}^T, \hat{\sigma}_1^2)^T$ maximize the following likelihood ¹¹

$$\begin{aligned} \ell &= \log L(Y|X) \\ &= \sum_{i=1}^n \sum_{t=0,1} \frac{1}{2} \left[I(T_i = t) \left\{ -\log(2\pi) - \log((\alpha_{t,2} + \alpha_{t,3}^T X_i)^2 + \sigma_t^2) - \frac{(Y_i - \alpha_{t,0} - \alpha_{t,1}^T X_i)^2}{(\alpha_{t,2} + \alpha_{t,3}^T X_i)^2 + \sigma_t^2} \right\} \right]. \end{aligned} \quad 12$$

According to the M-estimator property, we have [1](#)

$$\sqrt{n}(\widehat{\theta} - \theta) \xrightarrow{d} N\left(0, \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\}\right]^{-1} P_0\left\{\frac{\partial \psi}{\partial \theta} \frac{\partial \psi}{\partial \theta^T}\right\} \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\}\right]^{-1}\right), \quad [2](#)$$

where P_0 is the true mean and [3](#)

$$\begin{aligned} & \psi(T, X, Y; \theta) \\ = & \sum_{t=0,1} \frac{1}{2} \left[I(T=t) \left\{ -\log(2\pi) - \log((\alpha_{t,2} + \alpha_{t,3}^T X)^2 + \sigma_t^2) - \frac{(Y - \alpha_{t,0} - \alpha_{t,1}^T X)^2}{(\alpha_{t,2} + \alpha_{t,3}^T X)^2 + \sigma_t^2} \right\} \right]. \end{aligned} \quad [4](#)$$

Let [5](#)

$$m_B(X; \theta) = \Phi\left(\frac{(\alpha_{1,0} - \alpha_{0,0}) + (\alpha_{1,1} - \alpha_{0,1})^T X - c}{\sqrt{((\alpha_{1,2} - \alpha_{0,2}) + (\alpha_{1,3} - \alpha_{0,3})^T X)^2 + (\sigma_0^2 + \sigma_1^2)}}\right), \quad [6](#)$$

and [7](#)

$$m_H(X; \theta) = \Phi\left(\frac{(\alpha_{0,0} - \alpha_{1,0}) + (\alpha_{0,1} - \alpha_{1,1})^T X - c}{\sqrt{((\alpha_{0,2} - \alpha_{1,2}) + (\alpha_{0,3} - \alpha_{1,3})^T X)^2 + (\sigma_0^2 + \sigma_1^2)}}\right), \quad [8](#)$$

thus, $\widehat{\text{TBR}}_c(x) - \text{TBR}_c(x) = m_B(x; \widehat{\theta}) - m_B(x; \theta)$ and $\widehat{\text{THR}}_c(x) - \text{THR}_c(x) = m_H(x; \widehat{\theta}) - m_H(x; \theta)$. [9](#)

By the Delta-Method, we have [10](#)

$$\begin{aligned} \sqrt{n}(\widehat{\text{TBR}}(x) - \text{TBR}(x)) & \xrightarrow{d} N(0, \sigma_{cB}^2(x; \theta)), \quad [11](#) \\ \sqrt{n}(\widehat{\text{THR}}(x) - \text{THR}(x)) & \xrightarrow{d} N(0, \sigma_{cH}^2(x; \theta)), \end{aligned}$$

where [12](#)

$$\sigma_{cB}^2(x; \theta) = \frac{\partial}{\partial \theta^T} m_B(x; \theta) \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\} \right]^{-1} P_0\left\{\frac{\partial \psi}{\partial \theta} \frac{\partial \psi}{\partial \theta^T}\right\} \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\} \right]^{-1} \frac{\partial}{\partial \theta} m_B(x; \theta), \quad [13](#)$$

and [14](#)

$$\sigma_{cH}^2(x; \theta) = \frac{\partial}{\partial \theta^T} m_H(x; \theta) \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\} \right]^{-1} P_0\left\{\frac{\partial \psi}{\partial \theta} \frac{\partial \psi}{\partial \theta^T}\right\} \left[P_0\left\{\frac{\partial^2 \psi}{\partial \theta \partial \theta^T}\right\} \right]^{-1} \frac{\partial}{\partial \theta} m_H(x; \theta). \quad [15](#)$$

Appendix E Proof of Theorem 3¹

Proof. The estimator $\hat{\theta} = (\hat{\alpha}_{0,0}, \hat{\alpha}_{0,1}^T, \hat{\alpha}_{0,2}, \hat{\alpha}_{0,3}^T, \hat{\alpha}_{1,0}, \hat{\alpha}_{1,1}^T, \hat{\alpha}_{1,2}, \hat{\alpha}_{1,3}^T)^T$ maximize the following likelihood²

$$\ell = \log L(Y|X) = \sum_{i=1}^n \sum_{t=0,1} \left[I(T_i = t) \left\{ Y_i \log(G(X_i; \theta_t)) + (1 - Y_i) \log(1 - G(X_i; \theta_t)) \right\} \right], \quad 3$$

where⁴

$$G(X; \theta_t) = \Phi\left(\frac{\alpha_{t,0} + \alpha_{t,1}^T X}{\sqrt{1 + (\alpha_{t,2} + \alpha_{t,3}^T X)^2}}\right). \quad 5$$

According to the M-estimator property, we have⁶

$$\hat{\theta} - \theta = - \left[P_0 \left\{ \frac{\partial^2}{\partial \theta \partial \theta^T} \psi(T, X, Y; \theta) \right\} \right]^{-1} \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \psi(T_i, X_i, Y_i; \theta) + o_p(1/\sqrt{n}), \quad 7$$

where⁸

$$\psi(T, X, Y; \theta) = \sum_{t=0,1} \left[I(T = t) \left\{ Y \log(G(X; \theta_t)) + (1 - Y) \log(1 - G(X; \theta_t)) \right\} \right]. \quad 9$$

Let $m_B(X; \theta) = \Phi_b(\mu(x; \theta), \Sigma(x; \theta))$, and $m_H(X; \theta) = \Phi_h(\mu(x; \theta), \Sigma(x; \theta))$, we have $\widehat{\text{TBR}}(x) - \text{TBR}(x) = m_B(x; \hat{\theta}) - m_B(x; \theta)$ and $\widehat{\text{THR}}(x) - \text{THR}(x) = m_H(x; \hat{\theta}) - m_H(x; \theta)$.¹⁰

By the Delta-Method, we have¹¹

$$\begin{aligned} \sqrt{n}(\widehat{\text{TBR}}(x) - \text{TBR}(x)) &\xrightarrow{d} N(0, \sigma_{bB}^2(x; \theta)), \\ \sqrt{n}(\widehat{\text{THR}}(x) - \text{THR}(x)) &\xrightarrow{d} N(0, \sigma_{bH}^2(x; \theta)), \end{aligned} \quad 12$$

where¹³

$$\sigma_{bB}^2(x; \theta) = \frac{\partial}{\partial \theta^T} m_B(x; \theta) \left[P_0 \left\{ \frac{\partial^2 \psi}{\partial \theta \partial \theta^T} \right\} \right]^{-1} P_0 \left\{ \frac{\partial \psi}{\partial \theta} \frac{\partial \psi}{\partial \theta^T} \right\} \left[P_0 \left\{ \frac{\partial^2 \psi}{\partial \theta \partial \theta^T} \right\} \right]^{-1} \frac{\partial}{\partial \theta} m_B(x; \theta), \quad 14$$

and¹⁵

$$\sigma_{bH}^2(x; \theta) = \frac{\partial}{\partial \theta^T} m_H(x; \theta) \left[P_0 \left\{ \frac{\partial^2 \psi}{\partial \theta \partial \theta^T} \right\} \right]^{-1} P_0 \left\{ \frac{\partial \psi}{\partial \theta} \frac{\partial \psi}{\partial \theta^T} \right\} \left[P_0 \left\{ \frac{\partial^2 \psi}{\partial \theta \partial \theta^T} \right\} \right]^{-1} \frac{\partial}{\partial \theta} m_H(x; \theta). \quad 16$$

Appendix F Estimates of Parameters and Their Asymptotic Properties in Models (3) and (4)

The corresponding formulas of (F1), (F2), (F3) and (F4) for the general models are:

$$\begin{cases} \text{TBR}_c(x) &= \Phi\left(\frac{(g_1(x) - g_0(x)) - c}{\sqrt{(h_1(x) - h_0(x))^2 + \sigma_0^2 + \sigma_1^2}}\right), \\ \text{THR}_c(x) &= \Phi\left(\frac{(g_0(x) - g_1(x)) - c}{\sqrt{(h_0(x) - h_1(x))^2 + \sigma_0^2 + \sigma_1^2}}\right), \end{cases}$$

and

$$\begin{cases} \text{TBR}(x) &= \Phi_b(\tilde{\mu}(x), \tilde{\Sigma}(x)), \\ \text{THR}(x) &= \Phi_h(\tilde{\mu}(x), \tilde{\Sigma}(x)), \end{cases}$$

where

$$\begin{aligned} \tilde{\mu}(x) &= -(g_0(x), g_1(x)), \\ \tilde{\Sigma}(x) &= \begin{pmatrix} 1 + h_0^2(x) & h_0(x)h_1(x) \\ h_0(x)h_1(x) & 1 + h_1^2(x) \end{pmatrix}. \end{aligned}$$

In estimation, we first model $g_t(X)$ and $h_t(X)$ as $g_t(X; \alpha_{t,1})$ and $h_t(X; \alpha_{t,2})$. Also let $\psi(T, X, Y; \theta)$ denote the log-density function, where $\theta = (\alpha_{0,1}, \alpha_{0,2}, \sigma_0^2, \alpha_{1,1}, \alpha_{1,2}, \sigma_1^2)^T$ in the continuous case and $\theta = (\alpha_{0,1}, \alpha_{0,2}, \alpha_{1,1}, \alpha_{1,2})^T$ in the binary case. The estimation for θ can be obtained by maximizing $P_n[\psi(T, X, Y; \theta)]$, denote as $\hat{\theta}$. Then $\text{TBR}(x)$, $\text{THR}(x)$, $\text{TBR}_c(x)$ and $\text{THR}_c(x)$ can be estimated by:

$$\begin{cases} \widehat{\text{TBR}}_c(x) &= \Phi\left(\frac{(g_1(X; \hat{\alpha}_{1,1}) - g_0(X; \hat{\alpha}_{0,1})) - c}{\sqrt{(h_1(X; \hat{\alpha}_{1,2}) - h_0(X; \hat{\alpha}_{0,2}))^2 + \hat{\sigma}_0^2 + \hat{\sigma}_1^2}}\right), \\ \widehat{\text{THR}}_c(x) &= \Phi\left(\frac{(g_1(X; \hat{\alpha}_{0,1}) - g_0(X; \hat{\alpha}_{1,1})) - c}{\sqrt{(h_1(X; \hat{\alpha}_{0,2}) - h_0(X; \hat{\alpha}_{1,2}))^2 + \hat{\sigma}_0^2 + \hat{\sigma}_1^2}}\right), \\ \widehat{\text{TBR}}(x) &= \Phi_b(\tilde{\mu}(X; \hat{\theta}), \tilde{\Sigma}(X; \hat{\theta})), \\ \widehat{\text{THR}}(x) &= \Phi_h(\tilde{\mu}(X; \hat{\theta}), \tilde{\Sigma}(X; \hat{\theta})), \end{cases}$$

where

$$\tilde{\mu}(X; \hat{\theta}) = (-g_0(X; \hat{\alpha}_{0,1}), -g_1(X; \hat{\alpha}_{1,1})),$$

$$\tilde{\Sigma}(X; \hat{\theta}) = \begin{pmatrix} 1 + h_0^2(X; \hat{\alpha}_{0,2}) & h_0(X; \hat{\alpha}_{0,2})h_1(X; \hat{\alpha}_{1,2}) \\ h_0(X; \hat{\alpha}_{0,2})h_1(X; \hat{\alpha}_{1,2}) & 1 + h_1^2(X; \hat{\alpha}_{1,2}) \end{pmatrix}. \quad 1$$

We estimate the variances of $\widehat{\text{TBR}}_c(x)$, $\widehat{\text{THR}}_c(x)$, $\widehat{\text{TBR}}(x)$ and $\widehat{\text{THR}}(x)$ by the plug-in estimator respectively. 2

Appendix G Identification When U Depends on X 3

Theorem G.1. *Under the Assumption 2:* 4

(i) *When the outcome is continuous, if the following model (A7.6) holds for $t=0,1$* 5

$$\begin{cases} Y_t = g_t(X) + h_t(X)U + \epsilon_t, \epsilon_t \sim N(\mu_t, \sigma_t^2), \epsilon_t \perp (X, U, \epsilon_u), \\ U = W(X) + \epsilon_u, \epsilon_u \sim N(\mu_u, \sigma^2), \epsilon_u \perp X \end{cases} \quad 6 \quad (\text{A7.6})$$

then the Condition A in the Appendix B is sufficient to identify the joint distribution of 7

(Y_0, Y_1) *given X .* 8

(ii) *When the outcome is binary, if the following model (A7.7) holds for $t=0,1$* 9

$$\begin{cases} Y_t^* = g_t(X) + h_t(X)U + \epsilon_t, \epsilon_t \sim N(\mu_t, \sigma_t^2), \epsilon_t \perp (X, U, \epsilon_u), \\ Y_t = I(Y_t^* > 0), \\ U = W(X) + \epsilon_u, \epsilon_u \sim N(\mu_u, \sigma^2), \epsilon_u \perp X, \end{cases} \quad 10 \quad (\text{A7.7})$$

then the following Condition D is sufficient to identify the joint distribution of (Y_0, Y_1) given X . 11

Condition D. $(g_t(X) + W(X)h_t(X), h_t(X))$ *belongs to the family $(\mathcal{S}_1(X), \mathcal{S}_2(X))$ for* 12
 $t = 0, 1$, where

$$(\mathcal{S}_1(X), \mathcal{S}_2(X)) = \left\{ (S_1(X; \beta_1), S_2(X; \beta_2)) \mid (\beta_1, \beta_2) \in \mathcal{A}, \right. \\ \left. \forall (\beta_1^{(1)}, \beta_2^{(1)}) \neq (\beta_1^{(2)}, \beta_2^{(2)}) \in \mathcal{A}, \frac{S_1(X; \beta_1^{(1)})}{\sqrt{1 + S_2^2(X; \beta_2^{(1)})}} \neq \frac{S_1(X; \beta_1^{(2)})}{\sqrt{1 + S_2^2(X; \beta_2^{(2)})}} \right\}. \quad 13$$

Proof. 1

(i) Without loss of generality, we assume $\sigma^2 = 1$ since otherwise it can be absorbed into 2
 $h_t(X)$, $\mu_u = 0$ since otherwise it can be absorbed into $W(X)$ and assume $\mu_t = 0$ since
 otherwise it can be absorbed into $g_t(X)$. Also, we assume $h_t(0) > 0$ since otherwise we
 use $U^* = -U$ to replace U . By a little arrangement, we have

$$Y_t = (g_t(X) + h_t(X)W(X)) + h_t(X)\epsilon_u + \epsilon_t. \quad \text{3}$$

Thus, 4

$$Y \Big| (X, T = t) \sim N(g_t(X) + h_t(X)W(X), h_t^2(X) + \sigma_t^2). \quad \text{5}$$

Then $(g_t(X) + h_t(X)W(X))$ and $(h_t^2(X) + \sigma_t^2)$ can both be identified, so is $h_t(X)h'_t(X)$. 6
 Since $h_t(X)$ belongs to $\mathcal{S}(X)$, we can also identify $h_t(X)$ and σ_t^2 .

Note that 7

$$P(Y_0, Y_1 | X = x) = P\left((g_0(x) + h_0(x)W(x)) + h_0(x)\epsilon_u + \epsilon_0, (g_1(X) + h_1(X)W(X)) + h_1(X)\epsilon_u + \epsilon_1\right). \quad \text{8}$$

Thus, we can identify the joint distribution of (Y_0, Y_1) given X . 9

(ii) Without loss of generality, we can assume that ϵ_u follows a standard normal distribution. 10

Also, we assume $\mu_t = 0$ since otherwise it can be absorbed into $g_t(X)$, $\sigma_t^2 = 1$ since
 otherwise we can use $\tilde{Y}_t^* = Y_t^*/\sigma_t$ to replace Y_t^* and $h_t(0) > 0$ since otherwise we can
 use $U^* = -U$ to replace U . By a little arrangement, we have

$$\begin{aligned}
P(Y = 1|X, T = t) &= P(Y_t = 1|X) \\
&= P(g_t(X) + h_t(X)W(X) + h_t(X)\epsilon_u + \epsilon_t > 0|X) \\
&= \int \int \frac{1}{2\pi} \exp\left(-\frac{s_u^2 + s_t^2}{2}\right) I(g_t(X) + h_t(X)W(X) + h_t(X)s_u + s_t > 0) ds_t ds_u \\
&= \int \int_{-\infty}^{g_t(X) + h_t(X)W(X) + h_t(X)s_u} \frac{1}{2\pi} \exp\left(-\frac{s_u^2 + s_t^2}{2}\right) ds_t ds_u \\
&= \int \int_{-\infty}^0 \frac{1}{2\pi} \exp\left(-\frac{s_u^2 + (g_t(X) + h_t(X)W(X) + h_t(X)s_u + s_t)^2}{2}\right) ds_t ds_u \\
&= \int_{-\infty}^0 \int \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left((1 + h_t^2(X))(s_u + \frac{h_t(X)(g_t(X) + h_t(X)W(X) + s_t)}{1 + h_t^2(X)}\right)^2\right. \\
&\quad \left. + \frac{(g_t(X) + h_t(X)W(X) + s_t)^2}{1 + h_t^2(X)}\right) ds_u ds_t \\
&= \int_{-\infty}^0 \frac{1}{\sqrt{2\pi}\sqrt{1 + h_t^2(X)}} \exp\left(-\frac{1}{2} \frac{(s_t + g_t(X) + h_t(X)W(X))^2}{1 + h_t^2(X)}\right) ds_t \\
&= \Phi\left(\frac{g_t(X) + h_t(X)W(X)}{\sqrt{1 + h_t^2(X)}}\right).
\end{aligned}$$

1

Thus, if the Condition D is satisfied, we can identify $(g_t(X) + W(X)h_t(X), h_t(X))$. Let 2

$K_t(x, \epsilon_u) = g_t(x) + h_t(x)W(x) + h_t(x)\epsilon_u$, we have

$$\begin{aligned}
\text{TBR}(x) &= \{1 - \Phi(K_0(x, s))\} \Phi(K_1(x, s)) f_{\epsilon_u}(s) ds \\
&= \int \left\{ \int_{K_0(x, s)}^{\infty} \frac{1}{\sqrt{2\pi}} \exp(-s_0^2/2) ds_0 \right\} \left\{ \int_{-\infty}^{K_1(x, s)} \frac{1}{\sqrt{2\pi}} \exp(-s_1^2/2) ds_1 \right\} f_{\epsilon_u}(s) ds \\
&= \int \int_{-\infty}^0 \int_0^{\infty} \frac{1}{(2\pi)^{3/2}} \exp\left\{-\frac{(s_0 + K_0(x, s))^2 + (s_1 + K_1(x, s))^2 + s^2}{2}\right\} ds_0 ds_1 ds.
\end{aligned}$$

3

Let $K_{t,1}(x) = g_t(x) + h_t(x)W(x)$, thus $K_t(x, s) = K_{t,1}(x) + sh_t(x)$. Then the term in 4

$\exp\left(-\frac{1}{2}(\cdot)\right)$ can be arranged as **1**

$$\begin{aligned}
& \{s_0 + K_0(x, s)\}^2 + \{s_1 + K_1(x, s)\}^2 + s^2 \\
= & \{1 + h_0^2(x) + h_1^2(x)\}s^2 + 2\{(s_0 + K_{0,1}(x))h_0(x) + (s_1 + K_{1,1}(x))h_1(x)\}s \\
& + \{s_0 + K_{0,1}(x)\}^2 + \{s_1 + K_{1,1}(x)\}^2 \\
= & \{1 + h_0^2(x) + h_1^2(x)\}\left\{s + \frac{(s_0 + K_{0,1}(x))h_0(x) + (s_1 + K_{1,1}(x))h_1(x)}{1 + h_0^2(x) + h_1^2(x)}\right\}^2 \\
& + \frac{1}{1 + h_0^2(x) + h_1^2(x)}\left[\{s_0 + K_{0,1}(x)\}^2\{1 + h_1^2(x)\} + \{s_1 + K_{1,1}(x)\}^2\{1 + h_0^2(x)\}\right. \\
& \quad \left. - 2\{s_0 + K_{0,1}(x)\}h_0(x)\{s_1 + K_{1,1}(x)\}h_1(x)\right].
\end{aligned}$$

So **3**

$$\text{TBR}(x) = \int_0^\infty \int_{-\infty}^0 \frac{1}{(2\pi)S} \exp\left(-\frac{F}{2}\right) ds_0 ds_1, \quad \text{4}$$

where **5**

$$S^2 = 1 + h_0^2(x) + h_1^2(x), \quad \text{6}$$

$$\begin{aligned}
F &= \left[\{s_0 + K_{0,1}(x)\}^2\{1 + h_1^2(x)\} + \{s_1 + K_{1,1}(x)\}^2\{1 + h_0^2(x)\}\right. \\
&\quad \left.- 2\{s_0 + K_{0,1}(x)\}h_0(x)\{s_1 + K_{1,1}(x)\}h_1(x)\right]/S^2 \\
&= \left\{(s_0, s_1) - \mu\right\}\Sigma^{-1}\left\{(s_0, s_1) - \mu\right\}^T,
\end{aligned}$$

$$\begin{aligned}
\mu &= (-K_{0,1}(x), -K_{1,1}(x)), \quad \text{8} \\
\Sigma &= \begin{pmatrix} 1 + h_0^2(x) & h_0(x)h_1(x) \\ h_0(x)h_1(x) & 1 + h_1^2(x) \end{pmatrix}.
\end{aligned}$$

Thus, $\text{TBR}(x) = \Phi_2((0, \infty), (-\infty, 0); \mu, \Sigma)$, where $\Phi_2(A_0, A_1; \mu, \Sigma)$ is the distribution **9**
function of bivariate normal vector with mean μ , covariance matrix Σ and integral region
 $A_0 \times A_1$. Similarly, we can derive the form for $\text{THR}(x)$. Thus, we can identify the $\text{TBR}(x)$
and $\text{THR}(x)$, so the joint distribution of (Y_0, Y_1) given X are identifiable. $\square$

Appendix H Additional Tables **10**

Table 1: The true value, bias, average estimated standard error (ASE), empirical standard error (ESE) and 95% confidence interval (CI) coverage in continuous case. Every table cell contains two elements, which corresponds to the population TBR_c (first row in each cell) and THR_c (second row in each cell) ($c = 0.5$) respectively.

Distribution of U	true value	bias	ASE	ESE	95% CI coverage
Normal	0.501	-0.001	0.017	0.017	0.945
	0.397	-0.001	0.016	0.016	0.949
t(3)	0.500	-0.001	0.017	0.017	0.951
	0.396	0.002	0.016	0.016	0.948
t(10)	0.499	< 0.001	0.017	0.017	0.953
	0.395	0.002	0.016	0.016	0.939
$\chi^2(3)$	0.501	-0.001	0.017	0.016	0.955
	0.397	< 0.001	0.016	0.016	0.954
$\chi^2(10)$	0.502	-0.002	0.017	0.017	0.952
	0.397	< 0.001	0.016	0.016	0.956
P(3)	0.502	-0.002	0.017	0.017	0.951
	0.398	-0.002	0.016	0.016	0.943
P(10)	0.503	-3e-03	0.017	0.017	0.933
	0.397	-7e-04	0.016	0.016	0.942
B(0.5)	0.501	-8e-04	0.017	0.017	0.949
	0.395	6e-04	0.016	0.016	0.953

Table 2: Estimates, estimated standard deviation (SD) and p -value of parameters of the 1

Mind Study

2

	$t = 0$			$t = 1$		
	Estimate	SD	p -value	Estimate	SD	p -value
Gender	−0.656	0.275	0.017	−0.248	0.321	0.439
CVD	0.581	0.353	0.100	0.100	0.395	0.801
Age	1.075	0.202	< 0.001	0.500	0.231	0.030
DSST	−0.483	0.190	0.011	−0.652	0.231	0.005
Race	0.619	0.309	0.045	0.355	0.383	0.354
U	1.768	0.791	0.025	0.148	0.374	0.693
$UGender$	−1.916	0.480	< 0.001	−1.742	0.414	< 0.001
$UCVD$	−0.321	0.398	0.420	−1.669	0.506	0.001
$UAge$	1.280	0.513	0.013	2.090	0.435	< 0.001
$UDSST$	−1.166	0.313	< 0.001	−1.157	0.339	0.001
$URace$	1.729	0.390	< 0.001	2.239	0.479	< 0.001
σ_t^2	1.080	0.277	< 0.001	1.992	0.491	< 0.001